Use case UC1: Schedule an appointment

Primary Actor: Patient (Authorized clinic staff may do it on the behalf of the patient)

Stakeholders and Interests:

Patient: Wants accurate information, relatively short wait time, different options.

Healthcare professional: Wants accurate schedule without conflicting appointments.

Clinic: Wants accurate schedule with the professionals practicing at that clinic.

Preconditions:

Patient is identified and authenticated.

Postconditions:

Appointment is saved to the healthcare professional's schedule.

Appointment is saved to the customer's profile.

Main success scenario:

1. The patient specifies the healthcare professional, clinic and desired starting date (defaults to today's date and to family physician).
2. The system displays the available time slots.
3. The patient may select an available time slot and request an appointment.
4. The system creates the appointment and associates it with the patient, healthcare professional and the clinic.
5. The system confirms the appointment is booked and displays the details.

Extensions (failure scenario):

2.a No available time slots with the requested parameters:

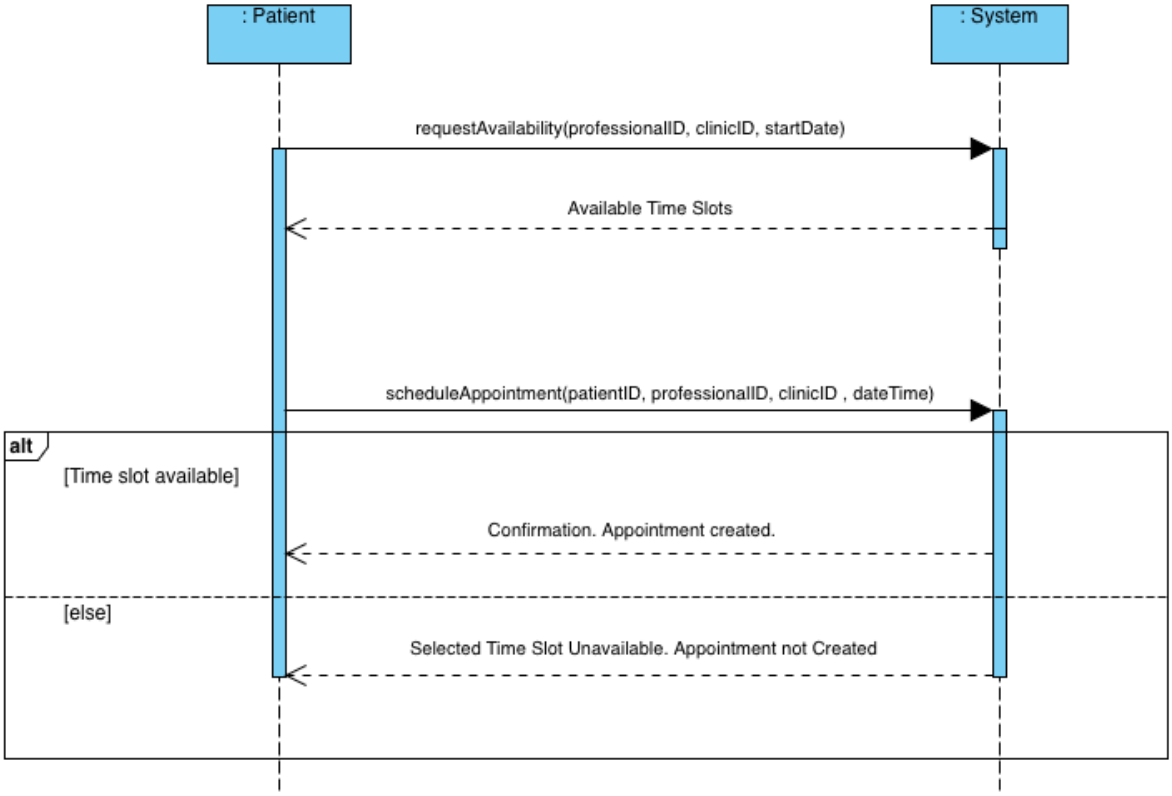
1. The system informs the patient and prompts the patient to change the parameters.
2. The patient changes the parameters and continues at step 1.

4.a The selected slot is no longer available:

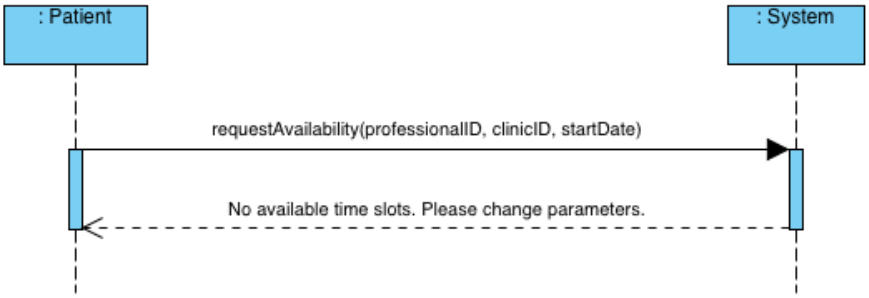
1. The system informs the patient the time slot is not available anymore and that the appointment was not created.
2. The system updates the availability, and the patient continues at step 3.

Special requirements: None.

Open issues: None.



SSD for Success Scenario & Failure Scenario 4.a



SSD for Failure Scenario 2.a

<u>System's Operation:</u>

requestAvailability(professionalID, clinicID, startDate)

scheduleAppointment(patientID, professionalID, clinicID, dateTime)

createReferral(referringProfessional, requestedSpeciality, requestedProfessional, clinicalReason, priority, referralStatus)
cancelAppointment(appointmentID)

Contract CO1: Request Availability

Operation: requestAvailability(professionalID, clinicID, startDate)

Preconditions:

The healthcare professional exists and works at that clinic.

The patient is authorized and identified.

Postconditions:

Nothing is modified in the domain.

Return:

Available time slots corresponding to the healthcare professional's schedule availability at that clinic, starting on the specified start date. An empty result represents no availability.

Contract CO2: Schedule an appointment (UC1)

Operation: scheduleAppointment(patientID, professionalID, clinicID, dateTime)

Preconditions:

The healthcare professional exists and works at that clinic.

The actor is identified and authorized to schedule an appointment for the patient.

Postconditions:

An Appointment instance **a** is created. (Instance Creation)

a was associated with the Patient. (Association formed)

a was associated with the Healthcare Professional. (Association formed)

a was associated with the Clinic. (Association formed)

Contract CO4: Make a referral (UC2)

Operation: createReferral(referringProfessional, requestedSpeciality, requestedProfessional, clinicalReason, priority, referralStatus)

Preconditions:

The referring healthcare professional is identified and authorized to issue referrals.

Patient exists.

Postconditions:

A referral instance **r** was created (Instance creation).

r was associated with associated with Patient (Association formed).

The referral was recorded by the system and sent to the patient with all information: referring professional, requested specialty, professional, clinical reason, priority, status (Instance creation).

The patient can make an appointment using the referral (Modification of an attribute)

Contract CO3: Cancel an appointment (UC3)

Operation: cancelAppointment(appointmentID)

Preconditions:

The actor is identified and authorized to cancel an appointment for the patient.

An appointment **a** identified by appointmentID exists.

A healthcare professional **p** linked to **a** exists.

Postconditions:

a.Status is changed to Cancelled. (Attribute modification)

a's time slot is removed from p.Schedule (Attribute modification)